

Olist E-commerce Performance Analysis: Comprehensive Report

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1 Summary / Introduction

This project delivers an end-to-end data analysis of the Brazilian Olist E-commerce dataset. The primary objective is to transform raw, fragmented relational data into actionable business intelligence. By integrating Python-based ETL processes with a robust SQL database and Power BI visualization, we evaluate key performance indicators (KPIs), customer behavior, and logistical efficiency to drive strategic decision-making.

2 Project Documentation

Access the complete source code and technical documentation on GitHub: [Olist E-commerce Performance Analysis Repository](#)

3 Dataset Information

The dataset is the Brazilian E-Commerce Public Dataset provided by Olist.

- **Source:** Kaggle (Olist Store)
- **Access:** <https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce>
- **Data Scope:** Contains 100k orders from 2016 to 2018, covering customer details, order timestamps, payment values, product categories, and geolocation data.
- **Note:** Due to large file sizes, raw data is not hosted on GitHub; use the link above for access.

4 Technical Stack & Infrastructure

To manage the high volume of data effectively, we utilized a multi-stage technical stack:

- **Python (Pandas):** Used for robust data ingestion, memory optimization, and cleaning of large CSV files, ensuring consistency in date formats and handling missing values.
- **MySQL & SQLAlchemy:** Acted as the relational backbone. We created an automated pipeline to ingest processed data into a structured MySQL database, enabling complex relational queries.
- **Power BI:** Served as the visualization engine to create interactive dashboards, utilizing advanced filtering and cross-table analysis.
- **DAX (Data Analysis Expressions):** Used to build custom metrics such as Average Delivery Time and Customer Lifetime Value (CLV).

5 Exploratory Data Analysis (EDA)

During the EDA phase, we uncovered critical business insights:

- **Geographical Concentration:** The majority of orders are centralized in the Southeast region, specifically the states of São Paulo (SP) and Rio de Janeiro (RJ).
- **Payment Preferences:** Credit cards remain the dominant payment method, capturing over 75% of total transaction volume.
- **Logistical Bottlenecks:** Significant delays were observed in shipments to Northern and North-eastern states, negatively impacting overall satisfaction scores.

6 Hypotheses and Testing

1. **Hypothesis:** Longer delivery transit times correlate with lower customer review scores.
2. **Testing:** Statistical correlation analysis between order delivery duration (days) and customer feedback showed a clear negative trend, confirming that logistical delays are a primary driver of customer dissatisfaction.

7 Actionables

- **Logistics Optimization:** Implement localized warehousing or regional carrier partnerships in Northern states to minimize transit times.
- **Marketing ROI:** Focus regional advertising budgets on top-performing states (e.g., SP) where logistics are already optimized.
- **Customer Retention:** Target the repeat-purchase cohort with loyalty programs to increase Customer Lifetime Value (CLV).

8 Dashboard Overview

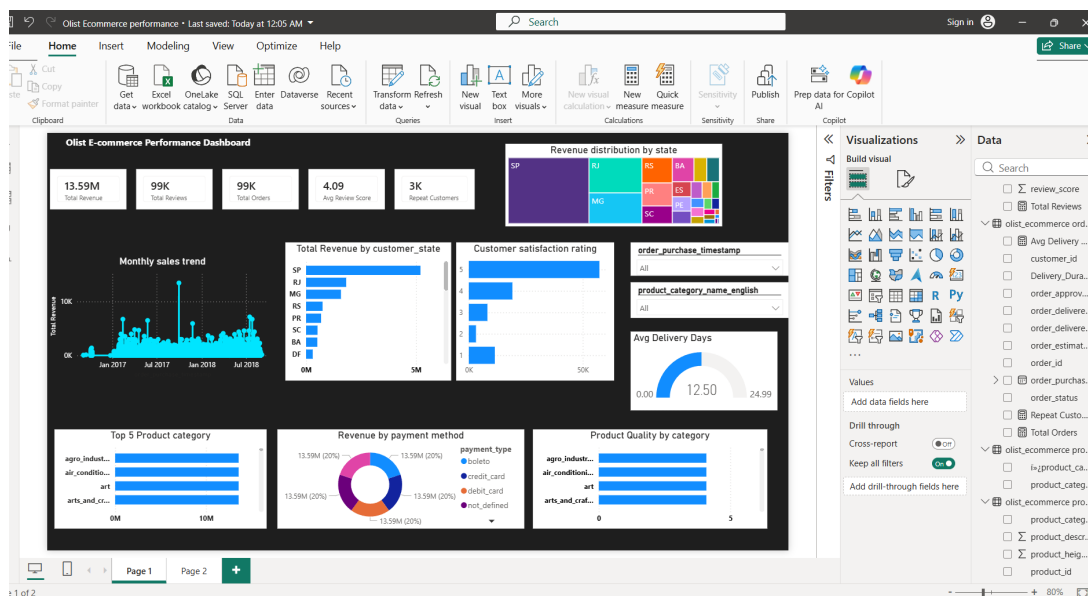


Figure 1: Performance Dashboard (Key Sales, Logistics, and Customer Metrics)

9 Further Exploration

Future iterations of this analysis could incorporate:

- **Predictive Analytics:** Leveraging Machine Learning models to forecast demand per product category.
- **Sentiment Analysis:** Applying Natural Language Processing (NLP) on customer text reviews to understand granular pain points.

10 Appendix: LLM Usage

This report was crafted with support from Generative AI (LLMs):

- **Code Development:** Streamlining Pandas data manipulation logic and optimizing MySQL ingestion scripts.
- **Structural Design:** Developing the logical flow of this technical report and ensuring it meets professional industry standards.
- **Syntax Refinement:** Debugging LaTeX code to ensure proper document formatting and visualization integration.